# OneSearch Pipeline —Review v4

*Enriched with learnings from Nespresso B2C (March 2026) and Danone Canada (Activia, Oikos, International Delight, Silk)*

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## What it is

A Python pipeline that answers one simple but structurally important question: for every keyword driving traffic to the site — through organic search, paid search, or both — what does the full picture look like? The output is a 57-column masterlist in Google Sheets, refreshed on demand by running a single script.

The Nespresso B2C implementation (March 2026) and the Danone Canada multi-brand deployment (Activia, Oikos, International Delight, Silk) provide two production references that reveal both the pipeline's strengths and the gaps that need to be addressed before it scales to portfolio clients.

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## Architecture — What is well designed

### Unified GSC × SQR spine

Building from keywords that have actually generated traffic (> 1 click) rather than a theoretical list avoids noise and grounds the analysis in real business data. The Nespresso dashboard confirmed this: 1,753 active keywords from the merged spine were sufficient to drive all scenario logic and recommendations.

### Jaccard trigram index

A pragmatic, performant fuzzy-matching solution with no external dependencies. The decision to iterate the 14,273 KS keywords through the small unified index (3,663 rows) rather than the reverse is well documented and technically sound — it prevents out-of-memory errors.

### Dual KS threshold (0.65 auto / 0.50–0.65 review)

Avoids forcing questionable matches into the masterlist while not discarding borderline ones. The human approval loop via approved = YES + re-run is clean.

### Source config via Google Sheet (REF_ID)

A solid architectural decision — it makes the tool multi-client without touching the Python code.

### Coverage metric (columns J/K)

The pipeline's most analytically valuable output. The Nespresso dashboard placed it at the center of every scenario view (coverage per subcategory, per period), confirming that clicks / estimated search volume is the single most useful ratio for a OneSearch diagnostic. The Danone SKILLSEOSEM formalises operational thresholds validated in production: < 3% invisible, 3–10% development, 10–30% competitive, > 30% domination.

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## Limitations & Points of Attention

### 1. The GA4 join via SE Ranking URL is the weakest link

Using SE Ranking as a bridge to attach GA4 conversions to keywords is structurally fragile: the join only works when SE Ranking has both matched the keyword and returned a landing page URL. Any keyword without that match gets no conversion data — not because conversions did not happen, but because the bridge is broken.

The correct methodology for SEO conversion attribution is to use GSC and GA4 directly:

1. 1. Export from GA4 all URLs that generated organic conversions.
2. 2. For each URL, pull from GSC the list of keywords that drove clicks to that URL, along with their click counts.
3. 3. Distribute the conversions across keywords pro rata to their click share on that URL (keyword conversions = total URL conversions × keyword clicks / total URL clicks).
4. 4. Round to the nearest integer (no decimals).

This gives keyword-level SEO conversions that are directly comparable to SEM conversions from Google Ads, without any dependency on SE Ranking URLs. This approach should replace the current GA4 join in a future version of the pipeline.

### 2. Blank Q4 SEM columns are a recurring operational risk

The current pipeline relies on a single SQR export in "comparison mode" (Q1 vs Q4) to populate both periods simultaneously. If the comparison period is missing or misconfigured, columns S, W, AA, and AJ are silently zero.

A more robust approach is to run two independent exports — one per period — and merge them in the pipeline:

- Export 1: SQR for Q1 2026 (Jan 1 – Mar 31)
- Export 2: SQR for Q4 2025 (Oct 1 – Dec 31)

This removes the dependency on Google Ads comparison mode, eliminates the risk of a blank period, and reflects the core value of the dashboard: period-over-period evolution is the primary analytical lens. The Danone Canada deployment already validates this approach — all four brands use separate Q1 and Q4 exports.

### 3. SE Ranking threshold at 0.50 is low, with no review safety net

Two corrections are needed here.

First, the threshold should be raised to 0.60. A Jaccard of 0.50 is too permissive for short or generic keywords and introduces false positives that silently corrupt position data in the masterlist.

Second, SE Ranking's role in the pipeline needs to be scoped correctly: SE Ranking should be used exclusively to attach a current SEO ranking position to each keyword — allowing GSC traffic data (clicks, impressions, CTR) to be read alongside its organic position at the time of the analysis. It should not enrich the masterlist with SE Ranking's own volume estimates, CPC, or intent labels. Keeping SE Ranking's contribution limited to the position column (column F) clarifies data lineage and reduces the impact of match errors.

### 4. No scenario or recommendation layer

The pipeline stops at the data layer. The Nespresso dashboard and the Danone SKILLSEOSEM.md demonstrate that the most client-facing value comes from the intelligence built on top of the masterlist: scenario classification (Domination SERP marque / Développement OneSearch / SEO à développer / SEA → Priorisation SEO), top-5 priority keywords per scenario, and coverage-driven recommendations. Both deployments use the same 4 scenarios and the same coverage thresholds — this logic is mature enough to be extracted into a reusable module.

**This is the biggest gap between the pipeline as documented and its actual deployment value.**

### 5. No multi-language handling

The pipeline hardcodes LANG = EN for Oikos USA. The Nespresso implementation required a 352-keyword EN→FR translation layer (KW_FR{}). For any French-market or bilingual client (including Danone Canada, which operates across English and French Canadian markets), this needs to be handled upstream in the pipeline normalization step.

### 6. No Quality Score (QS) integration

Both the Nespresso dashboard and the Danone SKILLSEOSEM.md treat Quality Score as a first-class signal: QS by keyword, QS by landing page, and the QS decision tree (< 5 = fix before raising bids, ≥ 7 = CPC reduction lever available). The current pipeline has no QS column. For any client where SEO/SEM overlap is significant, QS is essential for recommending where SEO investment reduces paid dependency.

### 7. SE Ranking position gaps — two-source fallback logic

A blank position in column F means the keyword has no ranking in the SE Ranking report — the site does not appear in the top 100. This should not be treated as missing data but as a signal in its own right.

However, a blank SE Ranking position does not mean the site has no organic presence on that keyword. GSC may still report an average position. The recommended fallback:

5. If column F (SE Ranking position) is blank, check GSC for an average position on that keyword.
6. If GSC returns a position, write it to column F rounded to the nearest integer (no decimals).

This two-source logic — SE Ranking first, GSC average as fallback — ensures maximum position coverage across the keyword spine without any SE Ranking crawl gap creating blind spots.

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## Additional observations — Danone Canada multi-brand deployment

### 8. Multi-brand architecture not supported

The pipeline is designed for a single brand per run (MASTER_ID hardcoded). The Danone Canada client operates four brands in parallel — Activia, Oikos, International Delight, Silk — each with its own masterlist and dashboard. There is no batch run logic, no cross-brand consolidated view, and no shared taxonomy across brands. For a portfolio client, this is a structural limitation that requires either a wrapper orchestration layer or a redesigned multi-tenant config.

### 9. The decision layer already exists — in SKILLSEOSEM.md

The Danone SKILLSEOSEM.md is a fully operational recommendation engine covering: three SEO position branches (1–3, 4–10, 11+), QS decision tree, ROAS arbitrage thresholds, seasonality triggers, hreflang cannibalization detection, and SQR gap analysis. The four Coverage scenarios are identical to those used in the Nespresso dashboard. This document is the missing "recommend.py" module referenced in point 4 — it simply needs to be codified.

### 10. Volume plan vs volume actif — critical distinction absent from the pipeline

The Danone SKILL formalises a distinction the pipeline does not document:

- volC — total volume of all keywords declared in the strategic plan
- osVolC — volume of keywords that actually generate OneSearch clicks

Confusing the two produces misleading period-over-period variations (e.g. −0.6% vs +3.4% on the same quarter). For Coverage calculation and trend analysis, only osVolC should be used. volC is reserved for sizing the total addressable market. This distinction must be documented in the pipeline column reference and enforced in the Coverage formula.

### 11. SQR filtering — Performance Max must be excluded

The pipeline ingests the SQR file without filtering by campaign type. The Danone SKILL documents a critical rule: filter to Search campaigns only, and exclude aggregate rows beginning with "Total:". Without this filter, Performance Max spend dilutes the Search ROAS, making SEM appear less efficient than it is (real-world example: Search ROAS 4.1× vs global ROAS collapsed when PMax is included). This filter must be applied during SQR normalization in norm_sqr().

### 12. GA4 event structure varies by client

Oikos USA uses mikmak_checkout and mikmak_click_offline_store events. Danone Canada uses standard GA4 Pages_and_screens exports (page views, not event-based conversions). The pipeline assumes a fixed GA4 format via ingest_ga4.py. For any new client, the GA4 ingestion layer must be configurable — the event name or metric column used as the conversion signal should be defined in the source config sheet, not hardcoded.

### 13. hreflang cannibalization — undetected blind spot

Danone operates multinational domains (activia.ca, oikos.ca, silkcanada.ca, internationaldelight.ca). The SKILL documents a full hreflang cannibalization scenario: a keyword where the URL ranking in the target country's SERP belongs to a different language or regional version of the site. The pipeline does not collect the ranked URL per country from SE Ranking, and therefore cannot detect this signal. For any multi-country client, this is a systematic blind spot that misattributes underperformance to content or SEO quality rather than a technical hreflang issue.

### 14. Connection-intent keywords pollute SEM performance metrics

The Danone SKILL identifies a class of keywords that generate SEM clicks with near-zero purchase intent: login, website, my account, mon compte, se connecter, espace client. These terms inflate click counts and depress ROAS without contributing to revenue. The pipeline currently includes them in the unified spine with no filter. They should be flagged or excluded during SQR normalization using a configurable exclusion list.

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